FPPF HYPERCOVERING GLUEING OF OBJECTS WITH VANISHING NEGATIVE SELF-EXTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The label lemma-fppf-glue-neg-ext-zero names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The counted proof is a complete sequential assembly; the first step cites the author's construction lemma lemma-construct-cartesian-objects-derived-modules, which is not itself a counted card (its proof omits one verification) - the dependency is recorded here and in the prerequisites rather than silently dropped.

Allowed standard prerequisites (editorial).

- Cartesian system from glueing data. Same chapter, \label{lemma-construct-cartesian-objects-derived-modules}. Line 7898: the K_n form the members of a simplicial system of the derived category of modules.
- Cartesian object from a cartesian system (own card). Same chapter, \label{lemma-cartesian-module-derived-from-simplicial} (own card L46301). Lines 7902-7904: produces K' with (K_n, K_phi) its deduced system.
- Unbounded fppf hypercovering descent for QCoh. Same chapter, \label{lemma-coh-descent-qcoh-unbounded-for-fppf-hypercovering} (registered card). Lines 7906-7908: realizes K' = La^*K for some K in D_QCoh(O_X).

Source reference. Original lemma 0GMF: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 3402–3402. Prerequisite pointer invoked at proof line(s) 7898; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0D9T: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 3539–3539. Prerequisite pointer invoked at proof line(s) 7902; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DHF: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 7676–7676. Prerequisite pointer invoked at proof line(s) 7906; it names a standard fact of the same chapter that the counted argument

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be an algebraic space over a scheme S. Let $a : U \to X$ be an fppf hypercovering. Suppose given $K_0 \in D_{QCoh}(U_0)$ and an isomorphism*

$$\alpha : L(f_{\delta_1^1})^* K_0 \longrightarrow L(f_{\delta_0^1})^* K_0$$

satisfying the cocycle condition on U_1. Set $\tau_i^n : [0] \to [n]$, $0 \mapsto i$ and set $K_n = Lf_{\tau_n^n}^* K_0$. Assume $\mathrm{Ext}^i_{\mathcal{O}_{U_n}}(K_n, K_n) = 0$ for $i < 0$. Then there exists an object $K \in D_{QCoh}(\mathcal{O}_X)$ and an isomorphism $La_0^* K \to K$ compatible with α.

Proof. The objects K_n form the members of a simplicial system of the derived category of modules by Lemma 0GMF. Then we obtain an object $K' \in D_{QCoh}(\mathcal{O}_{U_{étale}})$ such that (K_n, K_φ) is the system deduced from K', see Lemma 0D9T. Finally, we apply Lemma 0DHF to see that $K' = La^* K$ for some $K \in D_{QCoh}(\mathcal{O}_X)$ as desired. □